

Qingdao University of Technology
Qingdao, Shandong, China

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Editor-in-Chief

Concurrency and Computation: Practice and Experience

Dear Editor-in-Chief,

We submit the manuscript entitled “Concurrency-Aware Permission Decisions in Multi-Tenant LLM Agent Runtimes” for consideration as a research article in *Concurrency and Computation: Practice and Experience*.

The manuscript studies a concurrency failure in a multi-tenant agent runtime: tenant work can fill a bounded executor and delay a permission decision until its deadline, while a permissive timeout fallback converts executor unavailability into unsafe admission. We present Neuron, a Java 21 runtime architecture that reserves permission capacity and exports a resource-pressure signal to the escalation policy. The paper’s primary contribution is a systems characterization of executor contention, queueing, deadline semantics, and permission-verdict behavior; the security consequence is the measured failure mode rather than a claim of a new authorization theory.

The evaluation compares shared fail-open, shared fail-closed, isolated, and coupled architectures. It contains five independent JVM runs on two recorded environments, deadline and attacker-pressure surfaces, capacity sweeps, mixed load QoS, fixed API-trace replay, a process-separated Java permission service, and an additional executor-focused Windows replication. Raw JSONL observations, environment records, analysis scripts, generated values, and reproduction instructions are publicly archived at <https://github.com/dadawer-web/ouisani/tree/ccpe-repro-20260811/ccpe-repro-20260811>. The process-separated service is explicitly a same-host loopback experiment; we do not claim WAN, multi-node, GPU, or live-production behavior.

We believe the manuscript fits the journal’s emphasis on practical concurrent and distributed systems because it reports an implemented Java runtime, executor-level measurements, architecture baselines, and reproducible engineering trade-offs. The manuscript is original, is not under consideration elsewhere, and has been approved by both authors.

Thank you for your consideration.

Sincerely,

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